

# Question Paper

Exam Date & Time: 21-Nov-2025 (02:00 PM - 05:00 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: CSE 3124/CSE\_3124 COMPUTER NETWORKS

(SPL: COMPUTER SCIENCE & ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Explain the architecture of an electronic mail (e-mail) system with the help of a neat and labelled diagram. Clearly describe the main components involved-User Agents (UAs), Mail Transfer Agents (MTAs), and Mail Delivery Agents (MDAs)-and discuss the key protocols used for message transmission and retrieval, such as SMTP, POP3, and IMAP, highlighting their respective roles in the overall email communication process. (5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 1B) Describe the purpose of the three-way handshake in TCP connection establishment. What problem would occur if only a two-way handshake were used? (3)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 1C) Apply the OSI layering to a web transaction given an HTTP request from an application, identify the PDU name formed at each layer, and show how headers/trailers change the data size if Layer 4 adds 20 bytes, Layer 3 adds 20 bytes, and Layer 2 adds 18 bytes. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) Station A needs to send a message consisting of 10 packets to Station B using a sliding window of size 4. All packets are ready and can be transferred immediately. Selective repeat and Go-Back-N (GBN) are used at 2 different times, and every 5th packet gets lost for both protocols. (Ack's from B never lost). Let X and Y be the number of transmissions that A has to make in selective repeat and GBN, respectively, to ensure safe delivery to B. Then, what are the values of X and Y? (5)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2B) Analyze the behavior of the TCP Reno congestion control algorithm when a sender with  $cwnd = 12$  MSS and  $ssthresh = 12$  MSS receives three duplicate ACKs. Determine how Fast Retransmit and Fast Recovery modify the congestion window and threshold values: (3)
- Immediately after the triple duplicate ACK event.
  - After the next new ACK is received.
- Provide a step-by-step explanation of the  $cwnd$  and  $ssthresh$  updates and justify how Reno's mechanism ensures reliable transmission recovery.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 2C) Discuss how multiplexing and demultiplexing are achieved in the transport layer using port numbers. (2)

- 3A) Given a network topology as shown in Fig 3A, use Dijkstra's algorithm to determine the shortest path from A to all other nodes. (4)

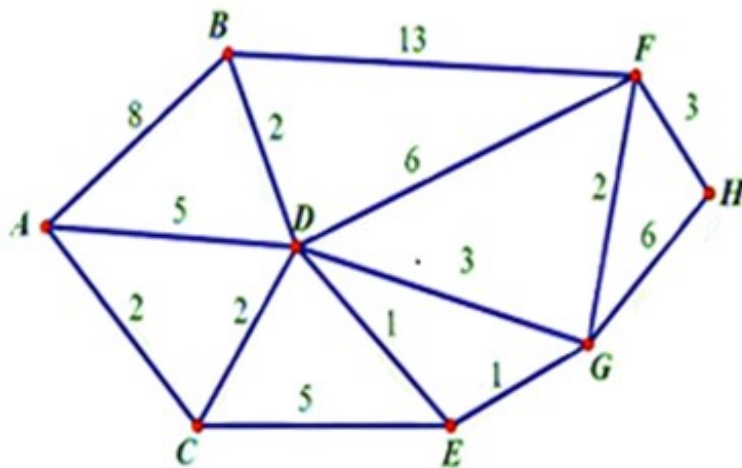


Fig3A

- 3B) Explain the difference between intra-AS and inter-AS routing on the Internet. Mention the main goals and give one example protocol for each. (3)

- 3C) Compare and contrast routing and forwarding in terms of their functions, scope, and role in the network layer. (3)

- 4A) An organization has been assigned the Class B network 172.16.0.0/16. (5)  
It must be subnetted to support five departments: Research, Administration, Faculty, Library, and IT Support, which require 500, 200, 100, 50, and 25 hosts respectively.  
Using Variable Length Subnet Masking (VLSM), design an efficient subnetting scheme that minimizes address wastage and allows room for future growth.  
Show the subnet addresses, masks, broadcast addresses, and usable host ranges for each subnet.  
Explain your steps clearly.

- 4B) What are some of the possible services that a link-layer protocol can offer to the network layer? (3)  
Which of these link-layer services have corresponding services in IP? In TCP?

- 4C) Analyze how input and output buffers contribute to the performance of a router. (2)

- 5A) Discuss the principles of the CSMA/CD (Carrier Sense Multiple Access with Collision Detection) protocol. In a bus network operating at 30 Mbps, if a collision occurs 20 microseconds after the first bit is transmitted, what should be the minimum frame length to ensure the sender detects the collision? (5)

5B) For a Message Frame : 1101011111 and Generator : 10011. compute CRC checksum and give the (3)  
transmitted stream.

Course Outcome: "CO5" / Cognitive Level: "Application"

5C) If all the links in the Internet were to provide reliable delivery service, would the TCP reliable (2)  
delivery service be redundant? Why or why not?

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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| Event Name | Subject |
|------------|---------|
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058-END SEMSTER-ODD-2025 CSE\_3124:Computer Networks

| #            | Course Outcome | Description | Marks     |
|--------------|----------------|-------------|-----------|
| 1            | 1              | CO 1        | 2         |
| 2            | 2              | CO 2        | 5         |
| 3            | 3              | CO 3        | 13        |
| 4            | 4              | CO 4        | 17        |
| 5            | 5              | CO 5        | 13        |
| <b>Total</b> |                |             | <b>50</b> |

\* **NA** - CO Not Selected

| Event Name | Subject |
|------------|---------|
|------------|---------|

058-END SEMSTER-ODD-2025 CSE\_3124:Computer Networks

| #            | Level       | No of Questions | Marks (%)        |
|--------------|-------------|-----------------|------------------|
| 1            | Remember    | 0               | 0(0%)            |
| 2            | Understand  | 1               | 3(6%)            |
| 3            | Application | 8               | 26(52%)          |
| 4            | Analysis    | 5               | 16(32%)          |
| 5            | Create      | 1               | 5(10%)           |
| 6            | Evaluate    | 0               | 0(0%)            |
| <b>Total</b> |             | <b>15</b>       | <b>50 (100%)</b> |